Chapter 1 : Overview of Information Retrieval

Adama Science and Technology University
School of Electrical Engineering and Computing
Department of CSE
Dr. Mesfin Abebe Haile (2024)

fppt.com

Information Retrieval

- Information retrieval (IR) is the process of finding *relevant documents* that satisfies *information need* of users from *large collections of unstructured text*.

- General Goal of *Information Retrieval*:
  - To *help users to find useful information* based on their *information needs* (with a minimum effort) despite:
    - *Increasing complexity* of Information,
    - *Changing needs* of user,
  - Provide *immediate random access* to the document collection.

fppt.com

Document Corpus

- *Large collections of documents* from *various sources*: news articles, research papers, books, digital libraries, Web pages, etc.

Sample *Statistics of Text Collections*:

- Dialog: (http://www.dialog.com/)
  - Claims to have *more than 20 terabytes of data in* > 600 Databases, > 1 Billion unique records.

- LEXIS/NEXIS: (http://www.lexisnexis.com/)
  - Claims *7 terabytes, 1.7 billion documents, 1.5 million subscribers*, 11,400 databases; > 200,000 searches per day; 9 mainframes, 300 Unix servers, 200 NT servers.

fppt.com

Document Corpus

- Large collections of documents from *various sources*: news articles, research papers, books, digital libraries, Web pages, etc.
- TREC (Text REtrieval Conference) collections:
  - It is *an annual information retrieval conference* & *competition*.
  - Total of about 10 GBs of *dataset of text for IR evaluation*.
- Web Search Engines:
  - *Google claim* to index *over 3 billion pages*.

fppt.com

Information Retrieval Systems ?

- Document (Web page) retrieval in *response to a query*.
  - Quite effective (at some things)
  - Commercially successful (some of them)
- But what goes on *behind the scenes*?
  - How do *they work*?
  - What *happens beyond the Web*?
- Web *search systems*:
  - Lycos, Excite, Yahoo, Google, Live, Northern Light, Teoma, HotBot, Baidu, …

fppt.com

Web Search Engines

- There are *more than 2,000 general web search engines*.
- The big four are *Google*, *Yahoo!*, *Live Search*, *Ask*.
  - *Scientific research* & *selected journals* search engine: Scirus, About.
  - *Meta search engine*: Search.com, Searchhippo, Searchthe.net, Windseek, Web-search, Webcrawler, Mamma, Ixquick, AllPlus, Fazzle, Jux2
  - *Multimedia search* engine: Blinkx
    - Visual search engine: Ujiko, Web Brain, RedZee, Kartoo, Mooter
    - Audio/sound search engine: Feedster, Findsounds
    - video search engine: YouTube, Trooker

–*Medical search engine*: Search Medica, Healia, Omnimedicalsearch,

fppt.com

Web Search Engines

- There are *more than 2,000 general web search engines*.
- The big four are *Google*, *Yahoo!*, *Live Search*, *Ask*.
  - *Index/Directory*: Sunsteam, Supercrawler, Thunderstone, Thenet1, Webworldindex, Smartlinks, Whatusee, Re-quest, DMOZ, Searchtheweb
  - *Others*: Lycos, Excite, Altavista, AOL Search, Intute, Accoona, Jayde, Hotbot, InfoMine, Slider, Selectsurf, Questfinder, Kazazz, Answers, Factbites, Alltheweb
- There are also *Virtual Libraries*: Pinakes, WWW Virtual Library, Digital-librarian, Librarians Internet Index.

fppt.com

Structure of an IR System

- An *Information Retrieval System* serves as a *bridge between* the *world of authors* and the *world of readers/users.*
  - Writers *present a set of ideas* in a document using a set of concepts.
  - Then Users seek the IR system for *relevant documents* that *satisfy their information need*.

- What is in *the Black Box*?
  - The black box is the *processing part* of the *information retrieval system*.

fppt.com

Information Retrieval vs. Data Retrieval

- Example of *data retrieval system* is a *relational database*.

	Data Retrieval	***Info Retrieval***
Data organization	Structured (Clear Semantics: Name, age…)	Unstructured (No fields (other than text)
Query Language	Artificial (defined, SQL)	Free text (“natural language”), Boolean
Query specification	Complete	Incomplete
Items wanted	Exact Matching	Partial & Best matching, Relevant
Accuracy	100 % (results are *always* “correct”)	< 50 %
Error response	Sensitive	Insensitive

fppt.com

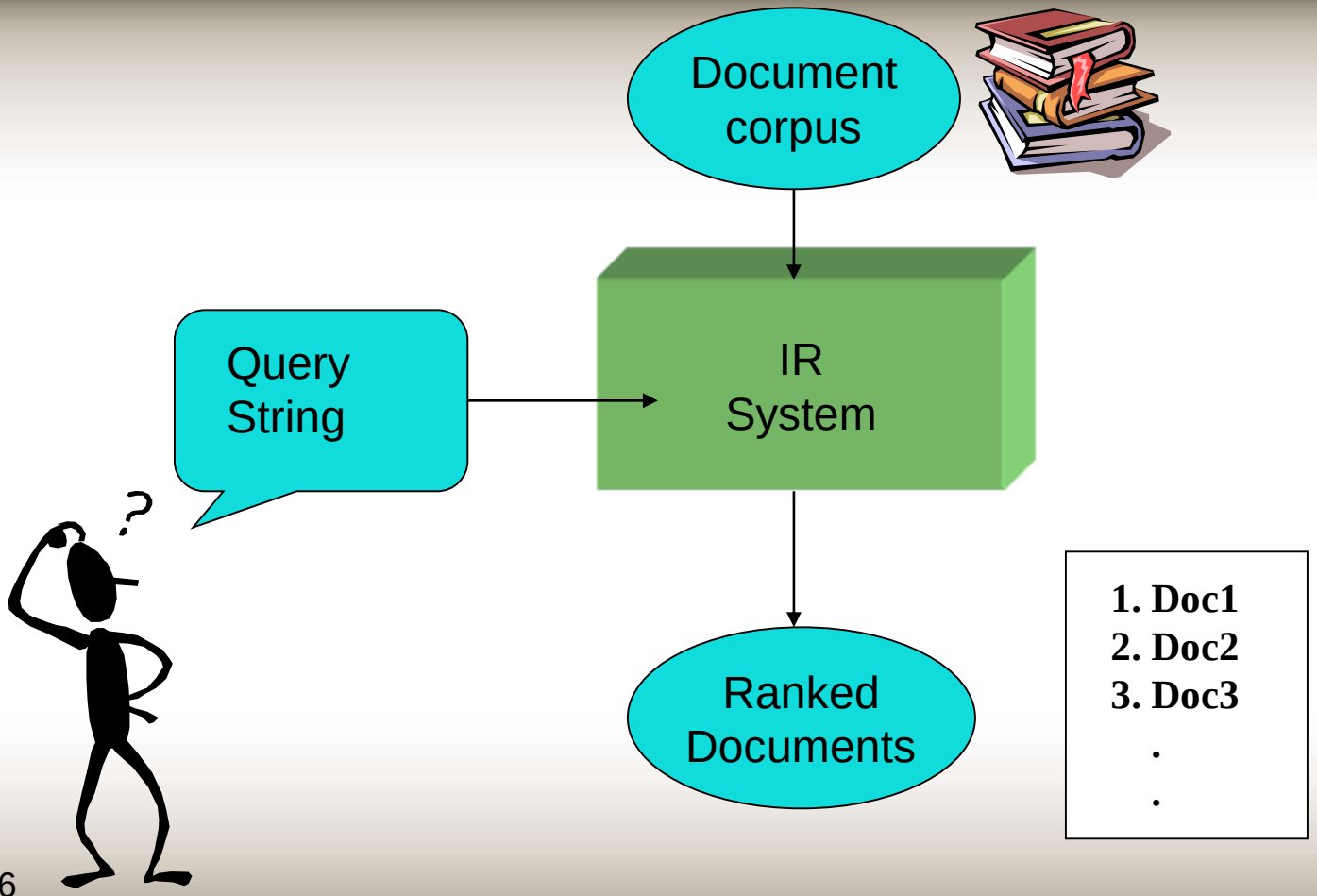
Typical IR Task

- Given:
  - A *corpus of document collections* (text, image, video, audio) published by *various authors*.
  - A *user information need* in the form of a *query*.

- An *IR system searches* for:
  - A *ranked set of documents* that are *relevant* to *satisfy information need of a user*.

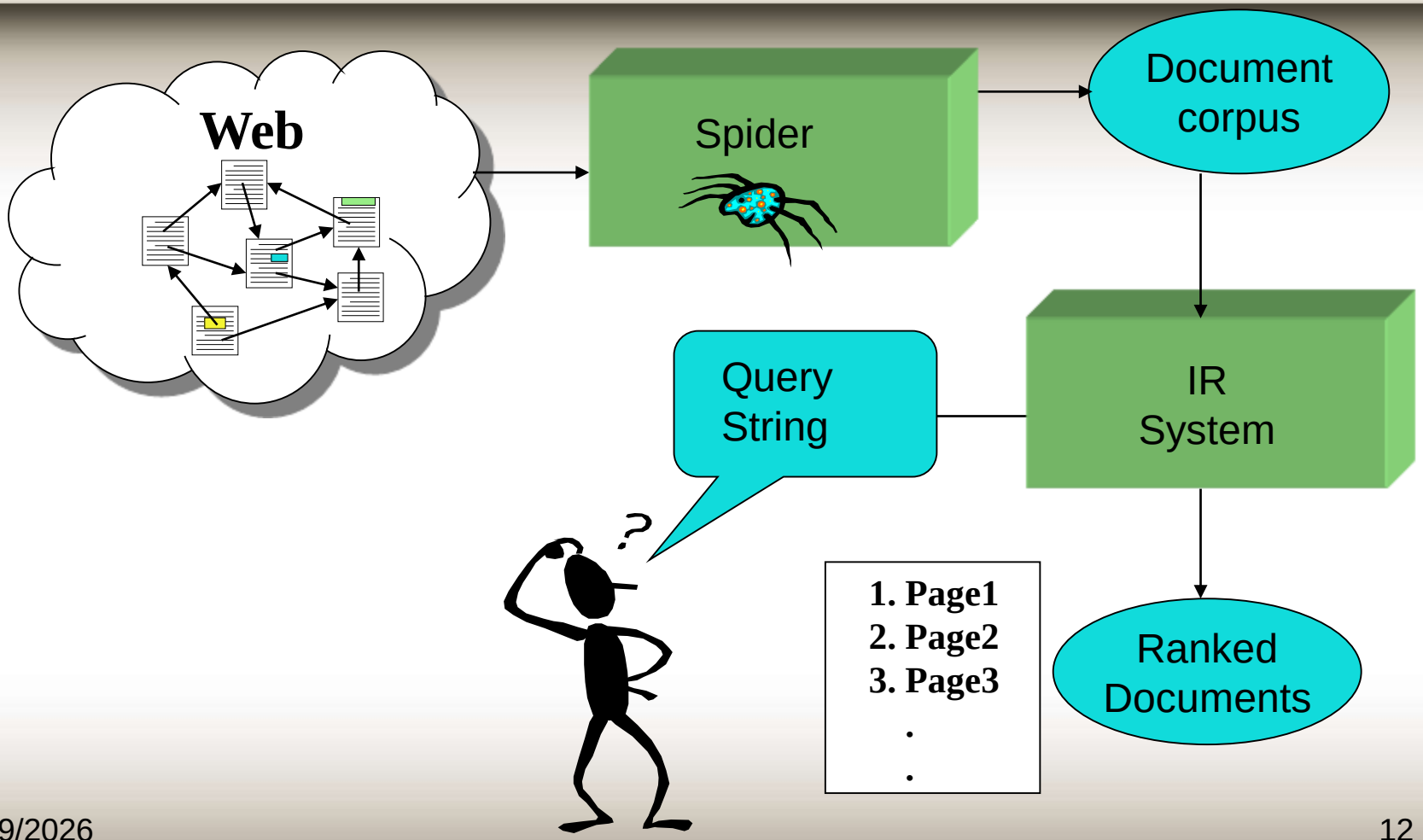
fppt.com

Typical IR System Architecture



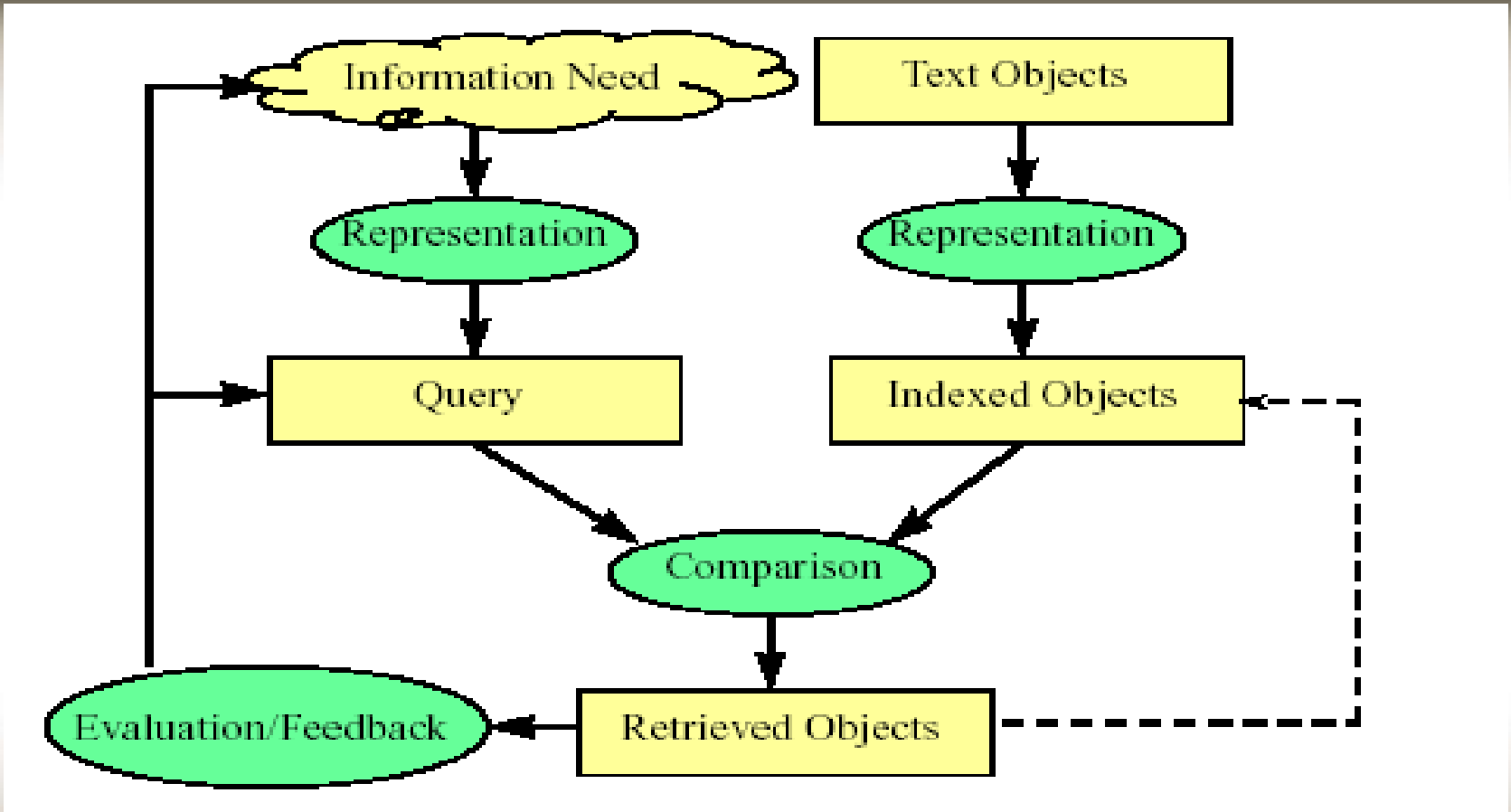
fppt.com

Web Search System



fppt.com

Overview of the Retrieval Process



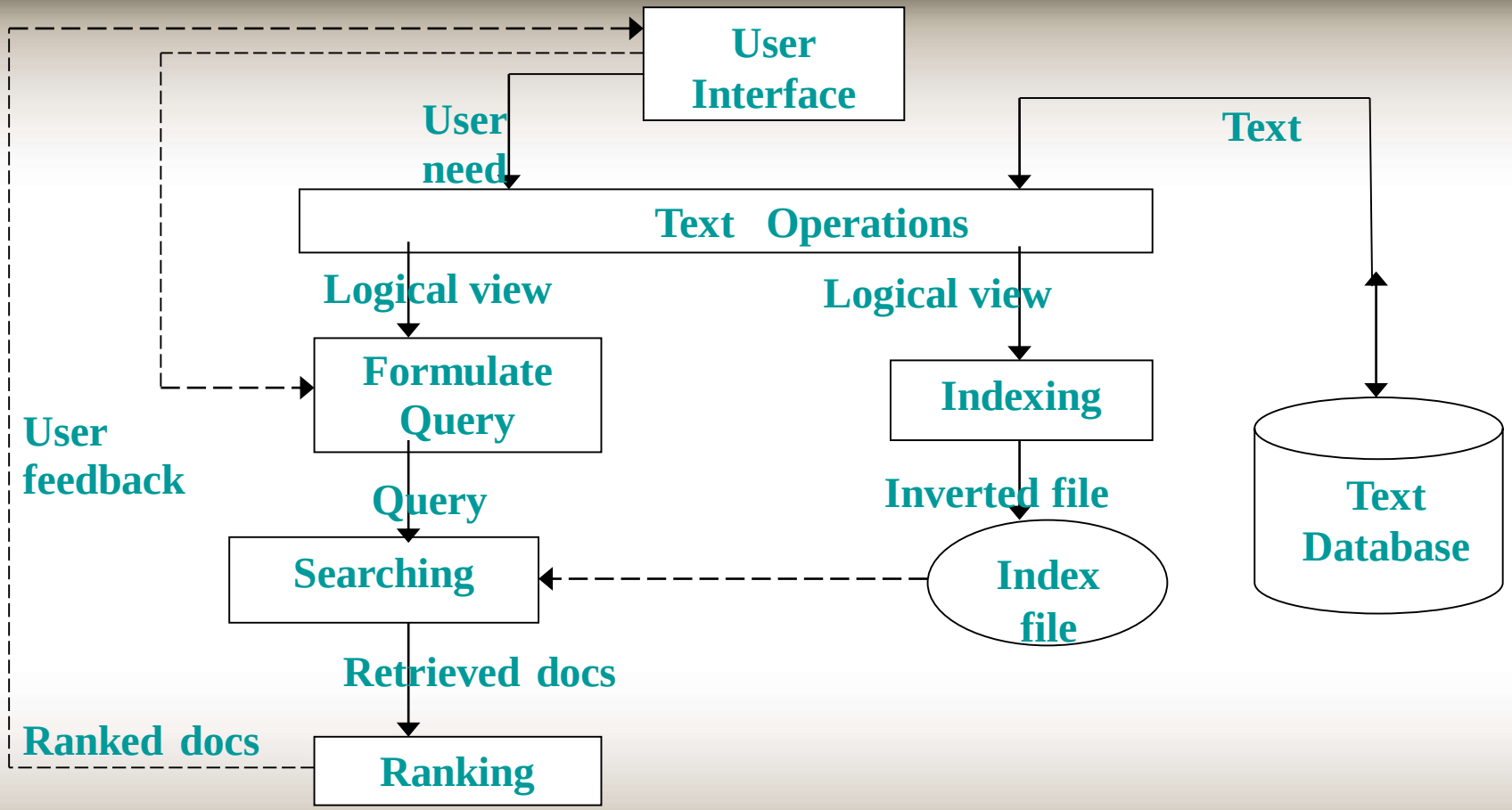
fppt.com

Issues that arise in IR

- Text *representation*:
  - What makes a "*good*" *representation*? The use of *free-text* or *content-bearing* index-terms?
  - How is a *representation generated* from text?
  - What are *retrievable objects* and how are they *organized*?
- Information *needs representation*:
  - What is an *appropriate query language*?
  - How can *interactive query formulation* and *refinement be supported*?
- *Comparing representations*:
  - What is a "*good*" *model* of retrieval?
  - How is *uncertainty represented*?
- *Evaluating effectiveness* of retrieval:
  - What are *good metrics*?
  - What constitutes a *good experimental test bed*?

fppt.com

Detail View of the Retrieval Process



fppt.com

Focus in IR System Design

- In improving *performance effectiveness* of the system.
  - *Effectiveness* of the system is evaluated in terms of *precision*, *recall*, …
  - Stemming, stop words, weighting schemes, matching algorithms.

- In improving *performance efficiency*. The concern here is:
  - Storage space usage, access time, …
  - Compression, data/file structures, space – time tradeoffs

fppt.com

Subsystems of IR system

- The *two subsystems* of an *IR system*:
- Indexing:
  - Is an *offline process of organizing documents* using *keywords extracted from the collection*.
  - Indexing is *used to speed up access to desired information* from document collection as per users query.

- Searching:
  - Is an *online process* that scans document corpus *to find relevant documents* that matches users query.

fppt.com

Statistical Properties of Text

- How is the *frequency* of *different words* distributed?
  - A few words are *very common*.
    - 2 most frequent words (e.g. “the”, “of”) can account for about *10% of word occurrences*.
  - Most words are *very rare*.
    - *Half the words* in a corpus appear *only once*, called “*read only once*”.

- How fast does *vocabulary size grow* with the *size of a corpus*?
  - Such factors affect the *performance of IR system* & can be used to select *suitable term weights* & *other aspects of the system*.

fppt.com

Text Operations

- Not all words in a document are *equally significant* to represent the *contents/meanings of a document*.
  - Some word carry *more meaning* than others.
  - Noun words are the *most representative* of a document content.
  - Therefore, need to *preprocess the text of a document* in a collection to be used as *index terms*.

- *Text operations* is the process of *text transformations* in to *logical representations*.
- It (text operation) generated a set of *index terms*.

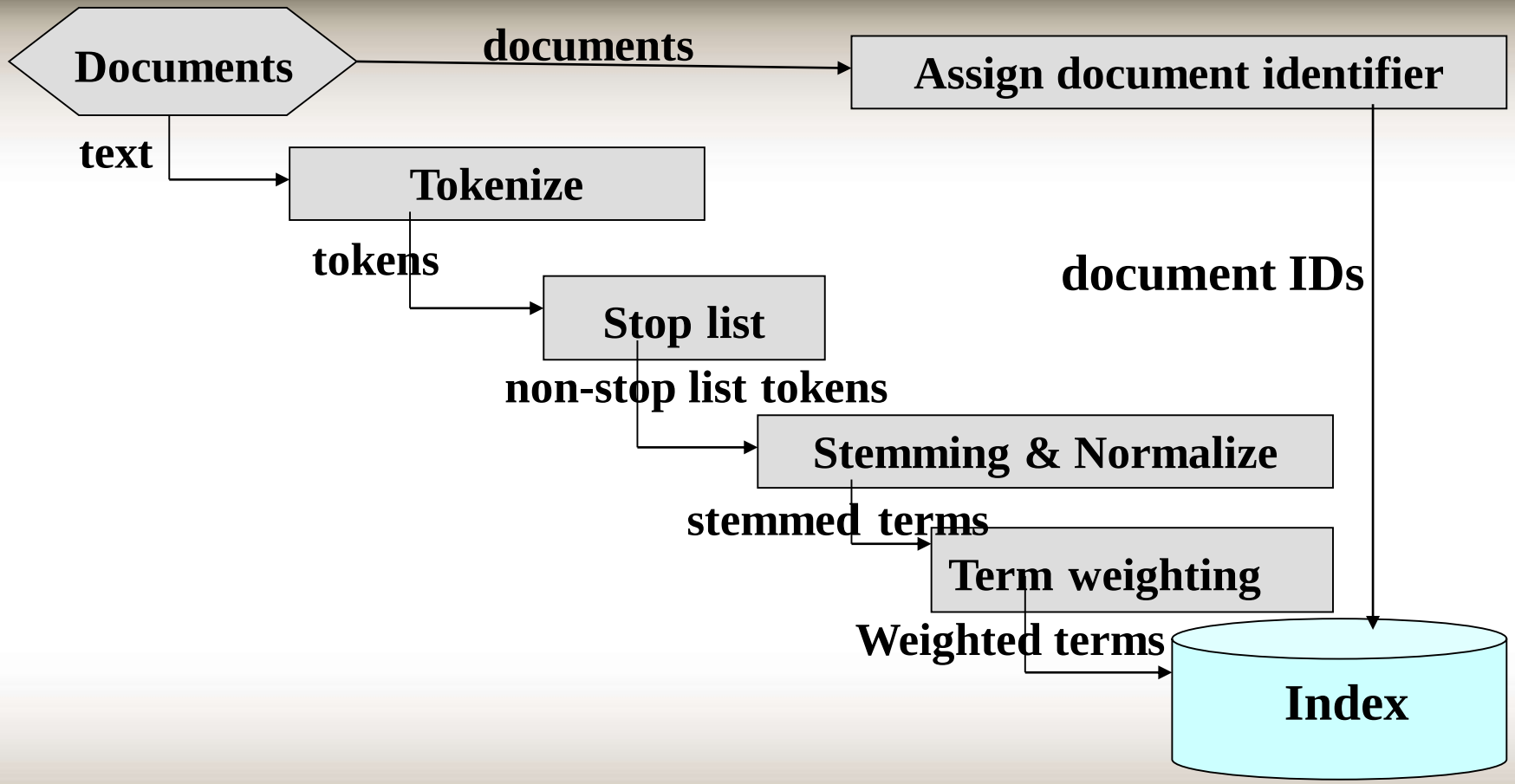
fppt.com

Text Operations

- Main operations for *selecting index terms*:
  - *Tokenization*: identify a set of words used to describe the content of text document.
  - *Stop words removal*: filter out frequently appearing words.
  - *Stemming words*: remove prefixes, infixes & suffixes.
  - *Design term categorization structures* (like thesaurus), which *captures relationship* for allowing the *expansion* of the original query with related terms.

fppt.com

Indexing Subsystem



fppt.com

Example: Indexing

Documents to be indexed.

Friends, Romans, countrymen.

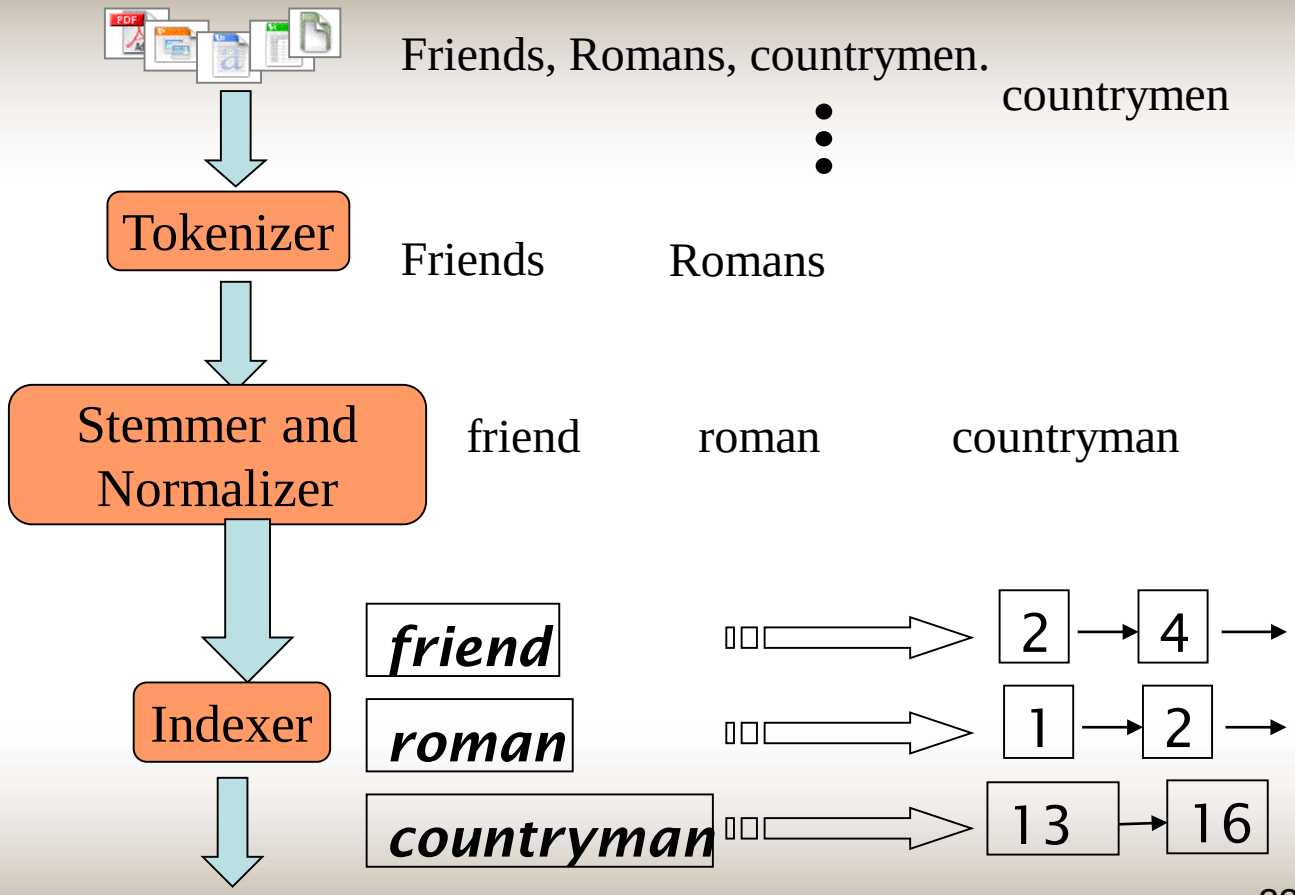
Token stream.

Friends Romans countrymen

Modified tokens.

friend roman countryman

Index File (Inverted file).



fppt.com

Index File

- An index file *consists of records*, called *index entries*.
- Index files are *much smaller than* the *original file*.
  - For 1 GB of TREC text collection the vocabulary *has a size of only 5 MB* (Ref: Baeza-Yates and Ribeiro-Neto, 2005)
  - This *size may be further reduced* by Linguistic *pre-processing* (like *stemming* & *other normalization* methods).
- The usual *unit for text indexing* is a *word*.
  - Index *terms - are* used to look up *records in a file*.
- Index file usually has *index terms* in a *sorted order*.
  - The *sort order of the terms* in the index file provides an order on a *physical file*.

fppt.com

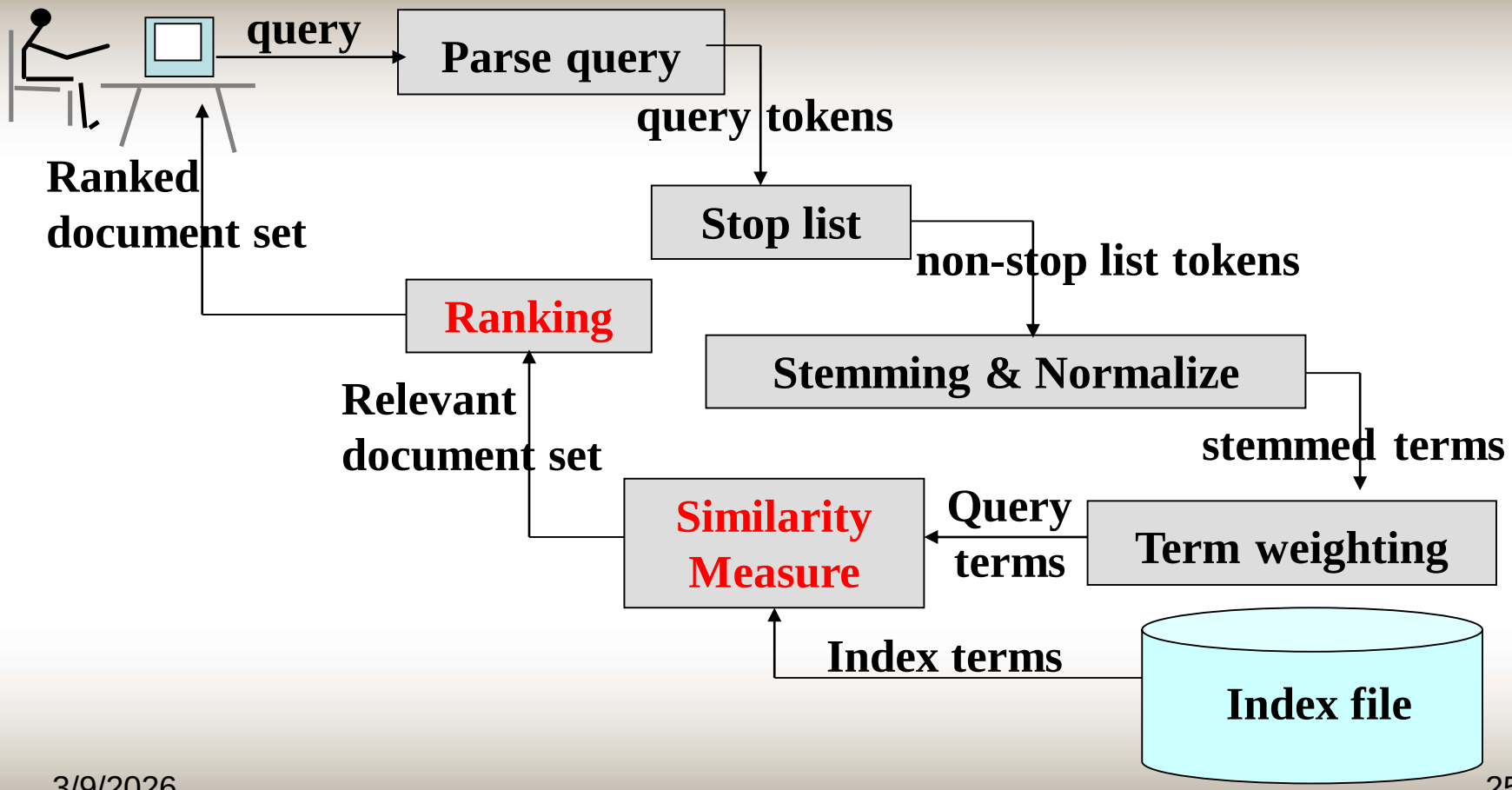
Building Index file

- *An index file* of a document is a file consisting of a *list of index terms* and a *link to one or more documents* that has *the index term*.
  - A *good index file maps* each keyword K_i to *a set of documents* D_i that *contain the keyword*.

- An index file is *list of search terms* that are organized for associative look-up, i.e., to *answer user's query*:
  - In which documents does a *specified search term appear*?
  - Where within each document *does each term appear*?
- For *organizing index file* for a collection of documents, there are *various options available*:
  - Decide what *data structure* and/or *file structure to use*. Is it *sequential file*, *inverted file*, *suffix array*, *signature file*, etc. ?

fppt.com

Searching Subsystem



fppt.com

IR Models - Basic Concepts

- One central problem regarding IR systems is the issue of predicting which documents are *relevant* and *which are not*.
  - Such a decision is usually dependent on a *ranking algorithm* which attempts to *establish a simple ordering* of the *documents retrieved*.
  - Documents appearning *at the top of this ordering* are considered to be *more likely to be relevant*.

- Thus *ranking algorithms* are at the *core of IR systems*.
  - The IR models determine the *predictions of what is relevant* and *what is not*, based on the notion of *relevance implemented* by the system.

fppt.com

IR Models - Basic Concepts

- After preprocessing, *N distinct terms remain* which are *Unique terms* that form the *VOCABULARY*.
- Let k_i be an *index term* i & d_j be a *document j*.
- Each *term*, i, in a *document* or *query j*, is given a *real-valued weight*, w_{ij}.
  - w_{ij} is a *weight associated* with (k_i,d_j). If $w_{ij} = 0$, it indicates that term does not *belong to document* d_j.
- The *weight* w_{ij} quantifies the importance of *the index term* for describing the *document contents*.
  - $vec(d_j) = (w_{1j}, w_{2j}, \ldots, w_{tj})$ is a *weighted vector associated* with the *document* d_j.

fppt.com

Mapping Documents & Queries

- Represent *both documents & queries* as N-dimensional vectors in a *term-document matrix*, which shows occurrence of terms in the *document collection/query*.
  - E.g. $\vec{d}_j = (t_{1,j}, t_{2,j}, \ldots, t_{N,j}); \vec{q}_k = (t_{1,k}, t_{2,k}, \ldots, t_{N,k})$
- An entry in *the matrix corresponds* to the "*weight*" of a term in the document; *zero means the term doesn't exist* in the document.

$$\begin{pmatrix} & T_1 & T_2 & \ldots & T_N \\ D_1 & w_{11} & w_{12} & \cdots & w_{1N} \\ D_2 & w_{21} & w_{22} & \cdots & w_{2N} \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ D_M & w_{M1} & w_{M2} & \cdots & w_{MN} \end{pmatrix}$$

- Document collection is mapped to *term-by-document matrix*.
- View as vector in *multidimensional space*.
  - Nearby *vectors* are *related*.

fppt.com

IR Models: Matching function

- IR models measure the *similarity between documents* and *queries*.
  - Matching function is a *mechanism used to match query* with a *set of documents*.
  - For example, the *vector space model* considers *documents* and *queries as vectors* in term-space and measures *the similarity of the document to the query*.
  - Techniques for matching include *dot-product*, *cosine similarity*, *dynamic programming*…

fppt.com

IR Models

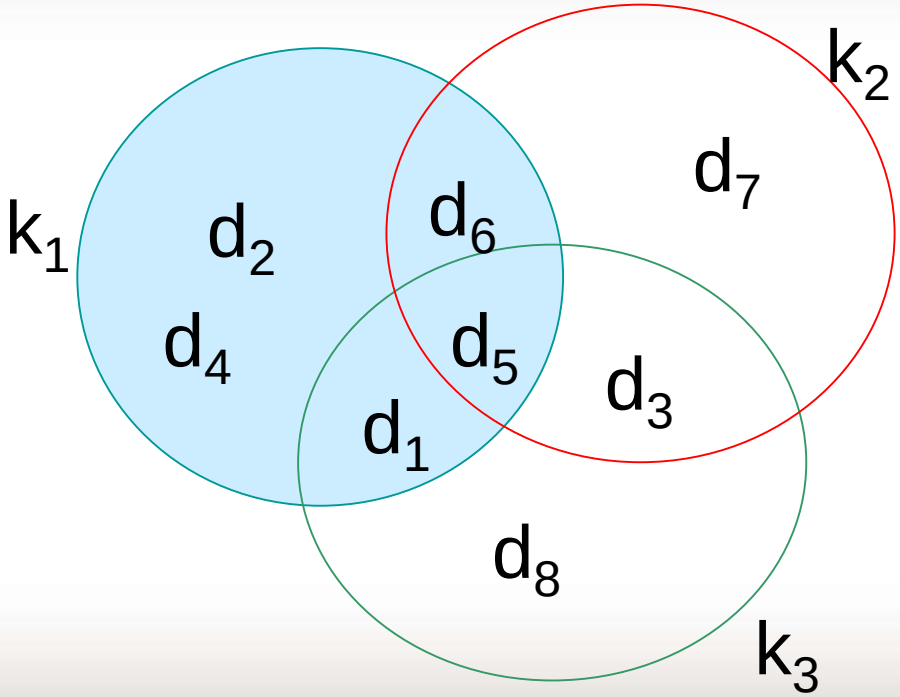
- A number of *major models* have been developed to *retrieve information*:
  - The *Boolean* model,
  - The *vector space* model,
  - The *probabilistic* model, and
  - Other models.

- *Boolean model*: is often referred to as the "*exact match*" *model*;
- Others are the "*best match*" *models*.

fppt.com

The Boolean Model: Example

- Generate the *relevant documents retrieved* by the *Boolean model* for *the query*:
  - $q = k_1 \wedge (k_2 \vee \neg k_3)$



fppt.com

IR System Evaluation?

- It provides the ability to *measure the difference between IR systems*.
  - How well do our *search engines work*?
  - Is system A *better than* B?
  - Under *what conditions*?
- *Evaluation drives* what to research:
  - Identify *techniques* that *work* and *do not work*,
  - There are many *retrieval models*/ *algorithms*/ *systems*.
    - Which one is *the best*?
- What is the *best method for*:
  - *Similarity measures* (dot-product, cosine, …)
  - *Index term selection* (stop-word removal, stemming…)
  - *Term weighting* (TF, TF-IDF,…)

fppt.com

Types of Evaluation Strategies

- *System-centered* studies:
  - Given *documents*, *queries*, and *relevance judgments*.
    - Try *several variations* of the system.
    - Measure which *system returns the "best" hit list*.

- *User-centered* studies:
  - Given *several users*, and *at least two retrieval systems*.
    - Have each user try the *same task on both systems*.
    - Measure which *system satisfy the "best" for users information need*.

fppt.com

Evaluation Criteria

- What are *some main measures* for evaluating an *IR system's performance*?

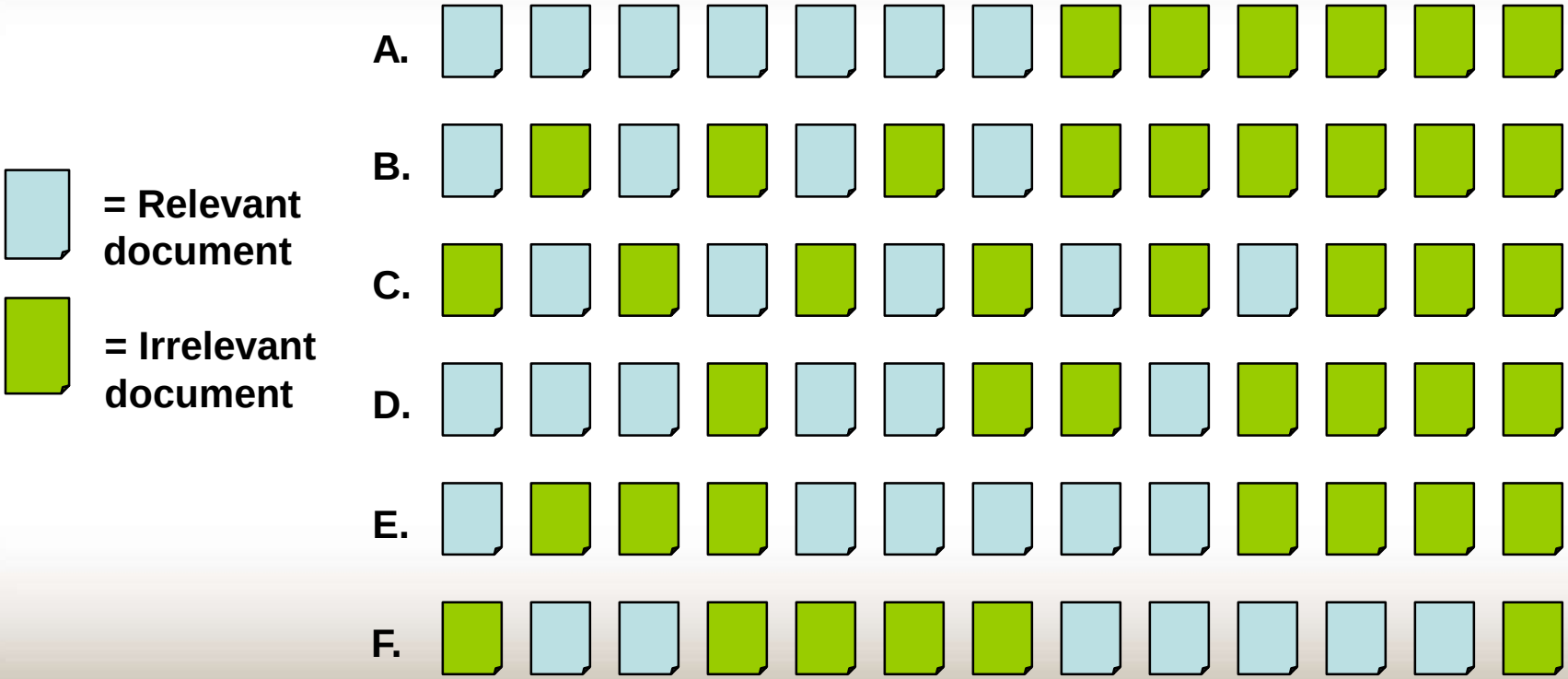
- Measure *effectiveness of the system*:
  - How is a *system capable of retrieving relevant documents* from the *collection*?
  - Is a *system better than another one*?
  - *User satisfaction*: How "good" are the documents that are returned as a *response to user query*?

  - "Relevance" of results *to meet information need of users*.

fppt.com

Retrieval scenario

- The *scenario where 13 results* retrieved by *different search engine* for a *given query*. Which *search engine* you *prefer*? Why?



fppt.com

Measuring Retrieval Effectiveness

- Metrics often used to *evaluate effectiveness of the system*.

	Relevant	Irrelevant
Retrieved	A	B
Not retrieved	C	D

- *Recall*:
  - Is percentage of *relevant documents retrieved* from the *database* in response to *users query*. (A / A + C)
- *Precision*:
  - Is percentage of *retrieved documents that are relevant* to the *query*. (A / A + B)

fppt.com

Query Language

- How *users query*?
  - The basic IR approach is *Keyword-based search*.
  - *Queries* are combinations of *words*.
- The *document collection* is searched for documents that contain *these words*.
- Word queries are intuitive, *easy to express* and *provide fast ranking*.
- There are *different query language*:
  - Single query,
  - Multiple query,
  - Boolean query, ... etc

fppt.com

Problems with Keywords

- May *not retrieve relevant documents* that include *Synonymous terms* (words with *similar meaning*).
  - "restaurant" vs. "café"
  - "Ethiopia" vs. "Abyssinia"
  - "Car" vs. "automobile"
  - "Buy" vs. "purchase"
  - "Movie" vs. "film"
- May retrieve *irrelevant documents* that include *Polysemy terms* (terms with *multiple meaning*).
  - "Apple" (company vs. fruit)
  - "Bit" (unit of data vs. act of eating)
  - "Bat" (baseball vs. mammal)
  - "Bank" (financial institution vs. river bank)

fppt.com

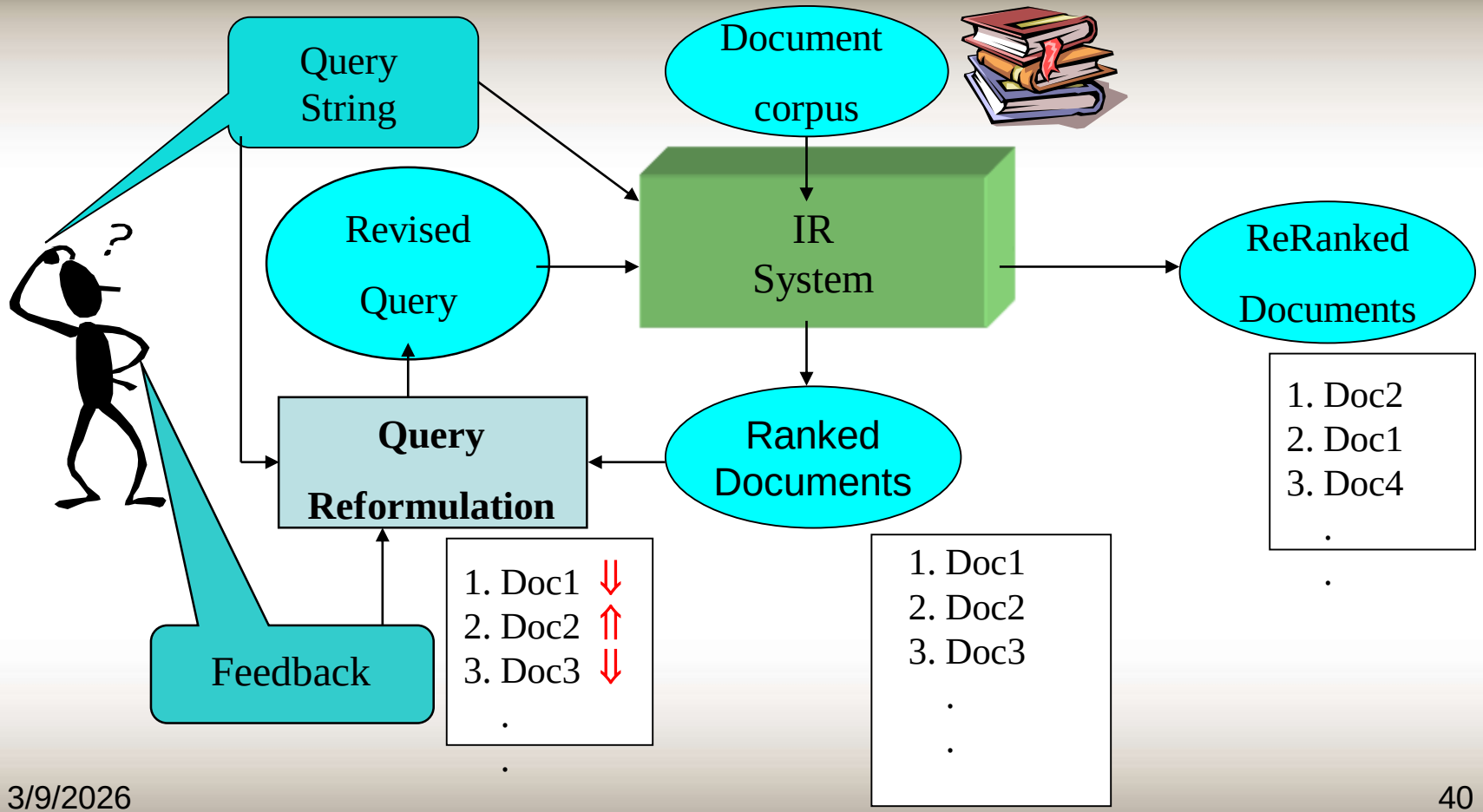
Relevance Feedback

- After *initial retrieval results are presented*, allow the user to *provide feedback on the relevance* of one or more of the *retrieved documents*.

- Use this *feedback information* to *reformulate the query*.
- Produce *new results* based on *reformulated query*.
- Allows *more interactive*, *multi-pass process*.
- *Relevance feedback* can be automated in such a way that it allows:
  - Users relevance feedback,
  - Pseudo relevance feedback.

fppt.com

Users Relevance Feedback Architecture



fppt.com

Challenges for IR researchers and practitioners

- *Technical challenge*: what tools should IR systems provide to allow *effective* and *efficient manipulation of information* within such diverse media of *text*, *image*, *video* and *audio*?

- *Interaction challenge*: what *features should IR systems provide* in order to support a wide variety of users in their search for *relevant information*.

- *Evaluation challenge*: how can we *measure effectiveness* of retrieval? which tools and features are *effective* and *usable*, given the increasing diversity of *end-users* and *information seeking situations*?

fppt.com

Assignments - One

- Pick *three of the following concept* (which is not taken by other students). Review literatures (books, articles & Internet) (concerning the *meaning*, *function*, *pros* and *cons & application* of the concept).

1. Information Retrieval
2. Search engine
3. Data retrieval
4. Cross language IR
5. Multilingual IR
6. Document image retrieval
7. Indexing
8. Tokenization
9. Stemming
10. Stop words
11. Normalization
12. Thesaurus
13. Searching
14. IR models
15. Term weighting
16. Similarity measurement
17. Retrieval effectiveness
18. Query language
19. Relevance feedback
20. Query Expansion

fppt.com

Question & Answer

fppt.com

Thank You !!!

fppt.com